

MooMe is a mobile monitoring & measurement IoT-AI platform transforming Rwanda's dairy sector. Piloted across **6 districts** in two milk sheds — Nyagatare (extensive) & Burera (intensive) — it equips **1,200 dairy farmers** with smart collars, sensors, and an AI-powered app in **Kinyarwanda & English** for real-time disease detection, heat monitoring, feed optimization & data-driven decisions.

6

DISTRICTS

1,200

FARMERS

132

TRAINERS

12

CHALLENGES

! PROBLEM

- Low milk yields: 5–10L/day vs 25–35L potential
- Late disease detection — high vet costs & mortality
- Feed costs = 60–70% of production expenses
- Missed heat detection & long calving intervals
- Livestock theft in extensive grazing systems
- No farm data — decisions based on guesswork
- Weak value chain (farmers ↔ processors)
- Limited extension access — no digital tools
- Climate stress with no adaptive systems

⚡ MOOME IOT-AI SOLUTION

Disease Detection

Temp & movement sensors flag illness to farmer + vet

Heat & Oestrus

Activity sensors detect heat; AI predicts insemination

Feed Optimization

AI ration formulation + intake sensors cut costs 20–40%

GPS Tracking

Real-time location, geo-fencing alerts against theft

Water Monitoring

Smart meters track herd intake, alert on drops

Farm Dashboard

Yield, health, breeding — all in one app

AI Advisory

Chatbot + Kinyarwanda tips & video tutorials

Credit Scoring

Farm data enables loans & insurance access

↑ PROJECTED IMPACT

+50%

Milk yield increase

-40%

Feed cost reduction

+30%

Farmer income

-35%

Waste reduction

EarlY

Disease detection lowers mortality

Real-time

Data-driven decisions

📍 PILOT REGIONS

NORTHERN MILK SHED

Burera, Musanze, Gicumbi

INTENSIVE DAIRY FARMING

EASTERN MILK SHED

Nyagatare, Gatsibo, Kayanza

EXTENSIVE DAIRY FARMING

Two contrasting ecosystems with different farming systems, cultures, and technology adoption mindsets — enabling robust platform validation.

🎓 TRAINING CASCADE (TOT)

12

Master Trainers

Data analysis, reporting & coaching. 1 per district → 12 facilitators + 100 farmers

120

Facilitators (Animal Scientists)

ToT program — 20/district. Field training, app handling, reporting

1,200

Model Dairy Farmers

Smartphone-equipped. Trained in Kinyarwanda & English on MooMe app

📦 DELIVERABLES

- 1 Baseline survey — AI-IoT perception & adoption
- 2 Trainer & farmer selection process
- 3 Device & sensor KIT procurement
- 4 Web & mobile app customization for Rwanda
- 5 Field testing, ToT validation & farm deploy
- 6 Data analysis, reporting & farmer feedback
- 7 Publication of research findings
- 8 Stakeholder workshop

18 Tablets

1,200 Collar KITs

1 Laptop

Business Plan & Implementation Strategy

📅 PROJECT TIMELINE — 24 MONTHS

PHASE 1

Months 1–3

Baseline survey, stakeholder workshop, trainer & farmer selection, ethics approval, procurement planning

PHASE 2

Months 4–6

Device & KIT procurement, web/mobile app design, AI model customization, platform setup for Rwanda

PHASE 3

Months 7–12

ToT training (12 Master + 120 Facilitators), field testing, validation, sensor calibration, app piloting

PHASE 4

Months 13–20

Full deployment to 1,200 farmers, ongoing mentoring, data collection, farm-level implementation

PHASE 5

Months 21–24

Data analysis, reporting, farmer feedback, publication, final stakeholder workshop, Phase 2 planning

💰 REVENUE & SUSTAINABILITY

Farmer Subscriptions

Monthly app subscription: \$2–5/farmer. Phase 2: FAO funds nationwide rollout subscriptions.

Device Sales & Leasing

Collar + sensor KIT sales to cooperatives & farmers with installment plans.

Data Analytics Services

Premium dashboards for processors (Inyange), vets & government agencies.

Training & Consulting

ToT programs licensed to other countries. Knowledge transfer services.

SmartFeed Integration

Bundled with WOREX feed plant — feed + monitoring as a package.

Detailed Budget Plan

#	BUDGET LINE ITEM	UNIT/QTY	UNIT COST (USD)	TOTAL (USD)
A. Personnel & Human Resources				
A1	Project Manager (full-time, 24 months)	24 months	\$2,000/mo	\$48,000
A2	Principal Investigator — Senior Animal Scientist	24 months	\$2,500/mo	\$60,000
A3	IoT/AI Technical Lead	24 months	\$2,200/mo	\$52,800
A4	Software Developer (Mobile & Web)	18 months	\$1,800/mo	\$32,400
A5	Data Analyst / AI Specialist	18 months	\$1,600/mo	\$28,800
A6	Livestock / Environment Expert	120 days	\$150/day	\$18,000
A7	Veterinarian Consultant (field support)	100 days	\$120/day	\$12,000
A8	MSc Research Student (stipend + research)	24 months	\$400/mo	\$9,600
A9	12 Master Trainers (training & coaching days)	12 × 60 days	\$200/day	\$144,000
A10	120 Facilitators/Animal Scientists (ToT + field)	120 × 30 days	\$100/day	\$360,000
A11	Administrative & Finance Officer	24 months	\$1,200/mo	\$28,800

#	Budget Line Item	Unit/Qty	Unit Cost (USD)	Total (USD)
A12	M&E Officer (Monitoring & Evaluation)	24 months	\$1,400/mo	\$33,600
Subtotal A — Personnel				\$828,000
B. Devices, Hardware & Equipment				
B1	IoT Smart Collars (GPS + temp + activity sensor)	1,200 units	\$120/unit	\$144,000
B2	Sensor KIT per farmer (rumen/ear tag + gateway)	1,200 kits	\$180/kit	\$216,000
B3	Android Tablets for Master Trainers	18 units	\$220/unit	\$3,960
B4	Laptop for MSc Student	1 unit	\$650	\$650
B5	Smart Water Meters (per pilot site)	24 units	\$250/unit	\$6,000
B6	Milk Quality Sensors (temperature + pH)	12 units	\$350/unit	\$4,200
B7	IoT Gateways / Connectivity Hubs (per sector)	30 units	\$300/unit	\$9,000
B8	Solar Charging Stations for devices	60 units	\$85/unit	\$5,100
B9	Spare parts, cables, protective cases	Lump sum	—	\$4,500
B10	Shipping, customs & import duties (18% VAT)	Lump sum	—	\$45,000
Subtotal B — Devices & Hardware				\$438,410
C. Software & Digital Platform Development				
C1	MooMe Mobile App development (Android + iOS, bilingual)	1 app	—	\$28,000
C2	Web Dashboard & Admin Portal development	1 platform	—	\$18,000
C3	AI Model development (disease, heat, feed, climate)	4 models	\$8,000/model	\$32,000
C4	AI Chatbot (Kinyarwanda NLP + English)	1 system	—	\$12,000
C5	Dataset collection, labeling & training data	Lump sum	—	\$8,000
C6	Cloud hosting & data storage (AWS/Azure, 24 months)	24 months	\$250/mo	\$6,000
C7	SMS/USSD gateway for farmer alerts	24 months	\$150/mo	\$3,600
C8	App maintenance & updates (Year 1 + 2)	24 months	\$350/mo	\$8,400
C9	Cybersecurity audit & data protection compliance	2 audits	\$3,000	\$6,000
Subtotal C — Software & Digital				\$122,000
D. Training & Capacity Building				
D1	Master Trainer ToT Workshop (5 days × 12 trainers)	12 pax × 5 days	\$100/day p.p.	\$6,000
D2	Facilitator ToT Training (10 days × 120 facilitators, 6 districts)	120 pax × 10 days	\$80/day p.p.	\$96,000
D3	Farmer Training Sessions (3 days × 1,200 farmers, per district)	1,200 pax × 3 days	\$15/day p.p.	\$54,000
D4	Training venue rental (rural district halls, 60 sessions)	60 days	\$100/day	\$6,000
D5	Training materials (manuals, handouts, folders per trainee)	1,332 sets	\$10/set	\$13,320
D6	Refreshments for training (tea + lunch)	1,332 pax × avg 5 days	\$5/day	\$33,300
D7	Farmer transport allowance (to/from training)	1,200 pax × 3 days	\$5/day	\$18,000
D8	Trainer per diem (accommodation + meals, rural)	132 pax × avg 20 days	\$80/day	\$211,200
D9	Video tutorial production (Kinyarwanda + English)	20 videos	\$500/video	\$10,000
Subtotal D — Training & Capacity Building				\$447,820
E. Research, Monitoring & Evaluation				
E1	Baseline survey design & ethics approval	1 survey	—	\$2,500
E2	Baseline survey — enumerators, data collection (1,200 farmers)	1,200 respondents	\$12/resp	\$14,400
E3	Endline survey (same scope as baseline)	1,200 respondents	\$12/resp	\$14,400
E4	Midterm evaluation & field data validation	6 districts × 5 days	\$300/day	\$9,000
E5	Data analysis software (STATA + Python tools)	Lump sum	—	\$1,500
E6	Farmer feedback exercise (6 district sessions)	6 sessions	\$800/session	\$4,800
E7	Research publication fees (2 peer-reviewed papers)	2 papers	\$1,800/paper	\$3,600
E8	Conference presentation & dissemination	2 conferences	\$2,000	\$4,000
Subtotal E — Research & M&E				\$54,200
F. Logistics, Transport & Field Operations				
F1	Vehicle hire — 4×4 for field visits (24 months)	24 months	\$1,200/mo	\$28,800
F2	Fuel & vehicle maintenance	24 months	\$400/mo	\$9,600
F3	Motorcycle hire for facilitators (district level)	6 motos × 12 months	\$200/mo	\$14,400
F4	Field visit per diem — project staff (district visits)	150 days	\$80/day	\$12,000
F5	Communication & airtime (staff + farmer data bundles)	24 months	\$350/mo	\$8,400
Subtotal F — Logistics & Transport				\$73,200
G. Workshops & Stakeholder Engagement				
G1	Inception Workshop — Kigali (RAB, UK, MINAGRI, FAO, RYAF, Co-ops, Districts)	1 day × 60 pax	\$50/pax + venue	\$4,500
G2	Conference venue rental — Kigali hotel	2 days	\$600/day	\$1,200
G3	Midterm Review Workshop	1 day × 40 pax	\$50/pax + venue	\$3,200
G4	Final Stakeholder Workshop & Results Dissemination	1 day × 80 pax	\$50/pax + venue	\$5,500
G5	District-level stakeholder meetings (6 districts × 2)	12 meetings	\$400/meeting	\$4,800
G6	Farmer cooperative engagement sessions	12 sessions	\$300/session	\$3,600
Subtotal G — Workshops & Stakeholder				\$22,800
H. Administration & Overhead				
H1	Office rent & utilities (WOREX project office)	24 months	\$300/mo	\$7,200
H2	Office supplies & stationery	24 months	\$100/mo	\$2,400
H3	Financial audit (annual, external)	2 audits	\$3,000	\$6,000
H4	Insurance — devices, equipment & liability	24 months	\$500/mo	\$12,000
H5	Banking, legal & regulatory fees	Lump sum	—	\$3,000
H6	Branding, printing & visibility materials	Lump sum	—	\$4,500
Subtotal H — Administration				\$35,100
I. Contingency & Risk Mitigation				

#	BUDGET LINE ITEM	UNIT/QTY	UNIT COST (USD)	TOTAL (USD)
II	Contingency fund (5% of total project cost)	5%	—	\$100,127
Subtotal I — Contingency				\$100,127
GRAND TOTAL — MooMe IoT-AI Platform Project (24 Months)				\$2,121,657



BUDGET ALLOCATION

- 39.0% Personnel & HR (\$828,000)
- 20.7% Devices & Hardware (\$438,410)
- 21.1% Training & Capacity (\$447,820)
- 5.8% Software & Digital (\$122,000)
- 2.6% Research & M&E (\$54,200)
- 3.5% Logistics & Transport (\$73,200)
- 1.1% Workshops (\$22,800)
- 1.7% Administration (\$35,100)
- 4.7% Contingency (\$100,127)



DIGITAL ECOSYSTEM

Farmers → Vets → Feed Suppliers → Milk Processors → Finance

Real-time data sharing across the value chain. Integrated with **SmartFeed Rwanda** for feed-to-milk optimization. Phase 2: FAO funds nationwide rollout.

PHASE 2 — NATIONAL SCALE-UP
FAO funds capacity building, farmer subscriptions & digital solution implementation to **all districts nationwide**.



GENDER & YOUTH INCLUSION

80% WOMEN & YOUTH 20%

Women and youth are primary targets — 80% of all 1,332 trainees. Building digital literacy and livestock management capacity across 6 districts.

PARTNERS

WOREX Ltd FAO Rwanda RAB Univ. of Kigali

MINAGRI RYAF Co-ops Districts